

## Task 26: Asteroid Collision

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### Problem Description

We are given an array `asteroids` of integers representing asteroids in a row. For each asteroid, the absolute value represents its size, and the sign represents its direction. Find out the state of the asteroids after all collisions.

### Java Solution

```
class Solution {
    public int[] asteroidCollision(int[] asteroids) {
        Stack<Integer> stack = new Stack<>();
        for (int ast : asteroids) {
            boolean exploded = false;
            while (!stack.isEmpty() && ast < 0 && stack.peek() > 0) {
                if (stack.peek() < -ast) {
                    stack.pop();
                    continue;
                } else if (stack.peek() == -ast) {
                    stack.pop();
                }
                exploded = true;
                break;
            }
            if (!exploded) {
                stack.push(ast);
            }
        }

        int[] res = new int[stack.size()];
        for (int i = res.length - 1; i >= 0; i--) {
            res[i] = stack.pop();
        }
        return res;
    }
}
```